

Cross-Cultural Social Acceptability Classification with Human Feedback

Zhichao Chen

Jialong Chen

Jon Paino

Abstract

Social acceptability classification aims to determine whether an utterance or interaction is appropriate in a given social context. This task is important for building safer and more culturally aware language models, but social acceptability is often culture-dependent: the same behavior may be acceptable in one setting and inappropriate in another. We study whether language models can adapt their judgments when given explicit cultural context and human-written cultural policies. We construct a balanced benchmark of 90 culturally dependent social interaction examples across three labels: *acceptable*, *depends*, and *unacceptable*. We evaluate GPT-4.1 mini and Llama-3.1-8B-Instruct under three prompting conditions: *culture_only*, *policy*, and *few_shot_policy*. Our results show that policy prompting improves both models and that few-shot policy prompting performs best overall. GPT-4.1 mini achieves the strongest performance, reaching 0.822 accuracy and 0.814 macro-F1. However, the *depends* class remains the hardest category for both models, showing that culturally grounded ambiguity remains a major challenge for social acceptability classification.

1 Introduction

Social acceptability classification (SAC) asks whether a given interaction is socially appropriate in context. The task is relevant to language models used for social advice, workplace communication, education, and interpersonal assistance, where inappropriate judgments can cause discomfort or harm. Existing models often rely on broad notions of harm, toxicity, or politeness, but these categories do not always capture cultural variation.

Social norms are highly context-dependent. For example, directly disagreeing with a supervisor may be normal in some workplace cultures but disrespectful in more hierarchical settings. Similarly, norms around gift-giving, privacy, family obliga-

tions, classroom behavior, gender interactions, and directness can differ substantially across communities. A model that treats social acceptability as universal may therefore misclassify culturally specific behavior.

The main research question of this project is:

Can human-defined cultural feedback improve model performance on culturally dependent social acceptability classification?

We first evaluate baseline model performance when the model receives only an interaction and a target cultural context. We then examine whether explicit human-written cultural policies help models adjust their predictions. Finally, we test whether adding a small number of labeled demonstrations further improves performance.

Our contributions are as follows¹:

- We build a balanced 90-example benchmark for cross-cultural social acceptability classification.
- We compare a proprietary model, GPT-4.1 mini, with an open-weight instruction-tuned model, Llama-3.1-8B-Instruct.
- We evaluate three prompting conditions: *culture_only*, *policy*, and *few_shot_policy*.
- We show that human-written cultural policies improve model performance, while context-dependent cases remain difficult.

2 Related Work and Model Selection

Social norms and acceptability. Forbes et al. (2020) introduce Social-Chem-101, a large corpus of 292k rules of thumb derived from everyday situations. Their work frames social norms

¹Our code is available at https://github.com/Chen-Zhichao/CS263_Proj.git.

as structured judgments over actions rather than a single toxicity or politeness score. The dataset includes dimensions such as social judgment, moral foundations, cultural pressure, and legality. However, Social-Chem-101 is primarily derived from English-speaking American annotators and does not condition each judgment on a target culture. We adopt a similar acceptability judgment framing while explicitly adding cultural context and policy explanations.

Cross-cultural evaluation. Rao et al. (2025) introduce NormAd, a benchmark for measuring the cultural adaptability of large language models. NormAd-Eti contains social-etiquette stories from many countries and evaluates models under different levels of cultural specificity, including country names, abstract values, and explicit rules of thumb. Their results show that even when the relevant social norm is provided, model accuracy remains below human performance. This motivates our smaller but more controlled benchmark, where each example includes a hand-written cultural policy.

Prompting and cultural alignment. Tao et al. (2024) evaluate cultural bias in GPT models using survey responses and show that cultural prompting can improve alignment for many countries and territories. Their method is lightweight because it changes inference-time prompts instead of requiring model retraining. Our project extends this idea from survey-value alignment to scenario-level acceptability classification by giving models explicit cultural policies before prediction.

Model selection. We selected GPT-4.1 mini and Llama-3.1-8B-Instruct as baseline models. GPT-4.1 mini is accessed through the OpenAI Responses API and represents a strong proprietary API model. Llama-3.1-8B-Instruct is accessed through an OpenAI-compatible endpoint using Hugging Face Inference Providers or a local vLLM server and represents an open-weight instruction-tuned model. We use a unified API wrapper so both models receive identical prompts and return structured JSON labels. This enables a fair comparison between a closed-weight API model and an open-weight instruction-tuned model.

3 Methodology

Dataset. We construct a benchmark of 90 culturally dependent interactions. The labels are bal-

anced, with 30 examples for each class: *acceptable*, *depends*, and *unacceptable*. Each entry contains: (1) an interaction, (2) a target cultural context, (3) a cultural value, (4) a human-written cultural policy, (5) a ground-truth label, and (6) a short explanation. The examples cover workplace hierarchy, classroom communication, politeness, privacy, family obligations, gender-related interactions, gift-giving, and direct versus indirect communication styles.

Prompting conditions. We evaluate each model under three settings. In the *culture_only* setting, the model receives the interaction and a target-culture identifier but no detailed policy. This measures whether the model can infer culturally appropriate behavior from pretrained knowledge alone. In the *policy* setting, the model receives the interaction, target culture, cultural value, and a human-written cultural policy before classification. This tests whether explicit cultural feedback helps the model adapt its judgment. In the *few_shot_policy* setting, the model receives the policy-based prompt plus three labeled examples as in-context demonstrations.

API implementation. Both models are called through a shared interface. The prompt asks the model to choose exactly one label from *acceptable*, *depends*, or *unacceptable*, and to return JSON containing a label and a brief rationale. The same dataset, prompt template, temperature setting, and output parser are used for both GPT-4.1 mini and Llama-3.1-8B-Instruct. Each example is evaluated independently.

Metrics. We report accuracy and macro-F1 on the full dataset. Accuracy measures the overall fraction of correct predictions. Macro-F1 averages F1 equally across the three labels, making it more sensitive to poor performance on the difficult *depends* class. We also report confusion matrices and analyze cases where the model disagrees with the ground-truth label.

4 Results

4.1 Overall Performance

Table 2 reports accuracy and macro-F1 for both models under all prompting conditions. Both models improve when provided with human-written cultural policies, and the few-shot policy setting achieves the best overall performance.

For Llama-3.1-8B-Instruct, accuracy increases from 0.589 under *culture_only* to 0.689 under

Field	Example
Interaction	During a visit to her friend’s house, Emily brought a beautifully wrapped gift. Upon receiving it, her friend immediately tore open the wrapping paper to see what was inside. Is what her friend did socially acceptable?
Cultural context	Tonga
Cultural value	Respect for the process, timing, and symbolism in gift-giving.
Ground-truth label	Unacceptable
Cultural policy	In Tongan gift-giving contexts, gifts may carry social and symbolic meaning, and recipients are often expected to show humility and respect rather than focusing too quickly on the material object. Opening a gift immediately in front of the giver may therefore be sensitive, especially if done eagerly or without acknowledging the giver’s gesture.
Explanation	Immediately tearing open the wrapping focuses attention too quickly on the material object and may fail to show humility or respect for the giver’s gesture. Because the action is described as immediate and eager, it is socially unacceptable in this context.

Table 1: Example data entry from our cross-cultural SAC benchmark.

Model	Condition	Acc.	Macro-F1
Llama-3.1-8B-Instruct	culture_only	0.589	0.473
Llama-3.1-8B-Instruct	policy	0.689	0.609
Llama-3.1-8B-Instruct	few_shot_policy	0.689	0.646
GPT-4.1 mini	culture_only	0.700	0.688
GPT-4.1 mini	policy	0.778	0.767
GPT-4.1 mini	few_shot_policy	0.822	0.814

Table 2: Accuracy and macro-F1 across models and prompting conditions.

policy, a gain of 10.0 percentage points. For GPT-4.1 mini, accuracy increases from 0.700 under culture_only to 0.778 under policy, a gain of 7.8 percentage points. Under few_shot_policy, GPT-4.1 mini reaches the best overall performance with 0.822 accuracy and 0.814 macro-F1.

4.2 Effect of Cultural Policies

The improvement from culture_only to policy suggests that models do not always infer culturally appropriate behavior from a culture identifier alone. When the prompt only contains the interaction and cultural context, the model must rely on pretrained cultural knowledge, which can lead to overgeneralized judgments. Adding a human-written policy provides an explicit norm that the model can apply to the scenario.

This effect is especially visible for Llama-3.1-

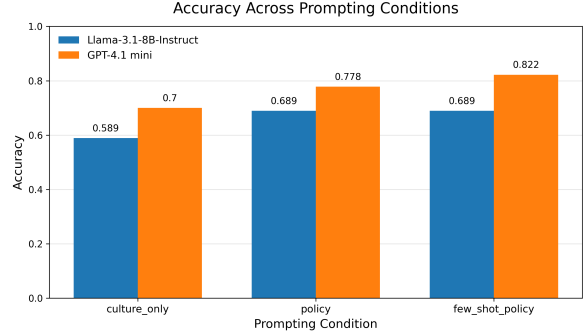


Figure 1: Accuracy across prompting conditions for Llama-3.1-8B-Instruct and GPT-4.1 mini. Policy prompting improves both models, and few-shot policy prompting achieves the best overall performance.

8B-Instruct, whose macro-F1 increases from 0.473 to 0.609 after adding policy information. This indicates that the policy condition improves not only the number of correct predictions, but also the balance of performance across labels.

4.3 Effect of Few-Shot Demonstrations

Few-shot demonstrations provide additional guidance. For GPT-4.1 mini, the few_shot_policy condition improves both accuracy and macro-F1 beyond the policy condition. For Llama-3.1-8B-Instruct, accuracy remains the same between policy and few_shot_policy, but macro-F1 increases from 0.609 to 0.646. This suggests that few-shot examples improve class balance even when they do not increase the total number of correct predictions.

Few-shot prompting is particularly useful because it demonstrates how the three labels should be used. The *depends* label is not simply a middle point between acceptable and unacceptable; it requires the model to recognize that the available information may be insufficient for a definitive judgment.

4.4 Confusion Matrix Analysis

Table 3 shows the confusion matrices for the few-shot policy condition. Rows are ground-truth labels and columns are model predictions.

For Llama-3.1-8B-Instruct, all acceptable examples are classified correctly, but the model struggles with context-dependent examples. Out of 30 *depends* examples, only 8 are classified correctly; 12 are predicted as acceptable and 10 are predicted as unacceptable. This suggests that Llama-3.1-8B-Instruct often collapses ambiguous cases into a binary judgment.

Llama-3.1-8B-Instruct			
GT	Pred. Acc.	Pred. Dep.	Pred. Unacc.
Acceptable	30	0	0
Depends	12	8	10
Unacceptable	5	1	24

GPT-4.1 mini			
GT	Pred. Acc.	Pred. Dep.	Pred. Unacc.
Acceptable	29	1	0
Depends	7	18	5
Unacceptable	1	2	27

Table 3: Confusion matrices for the few-shot policy condition. Rows represent ground-truth labels and columns represent predictions.

Condition	Llama	GPT
Culture only	0	23
Policy	5	22
Few-shot policy	9	21

Table 4: Number of predictions assigned to the *depends* class under each prompting condition. Here, Llama refers to Llama-3.1-8B-Instruct and GPT refers to GPT-4.1 mini.

For GPT-4.1 mini, the confusion matrix is more balanced. The model correctly classifies 18 out of 30 *depends* examples and performs strongly on both acceptable and unacceptable examples. However, even GPT-4.1 mini still misclassifies 12 *depends* examples, confirming that context-dependent social reasoning remains difficult.

4.5 The Difficulty of the Depends Class

The *depends* class is the hardest category for both models. This class requires the model to identify when a social judgment depends on missing details such as tone, relationship closeness, prior consent, formality, timing, or local expectations. In other words, the model must recognize uncertainty rather than force every example into a clear acceptable or unacceptable judgment.

Table 4 reports the number of predictions assigned to the *depends* class under each prompting condition. Under *culture_only*, Llama-3.1-8B-Instruct never predicts *depends*, showing a strong tendency toward binary classification. With policy and few-shot prompting, it begins to use the label more often. GPT-4.1 mini uses the *depends* label more consistently across all conditions.

5 Discussion

Human-written policies improve cultural judgment. Our results show that explicit cultural policies improve performance for both models. This supports the idea that lightweight inference-time feedback can help models adapt to cultural norms without retraining. A culture identifier alone may be too broad, while a policy gives the model a specific norm to apply.

Few-shot examples help define label boundaries. Few-shot prompting improves the best model performance and increases macro-F1. The demonstrations are useful because they show not only what the correct label is, but also how to reason from a cultural policy to a judgment. This is especially important for *depends*, where the correct answer often requires recognizing ambiguity.

GPT-4.1 mini outperforms Llama-3.1-8B-Instruct. GPT-4.1 mini outperforms Llama-3.1-8B-Instruct across all prompting conditions. It has a stronger baseline under *culture_only*, benefits from policy prompting, and achieves the best final result under *few_shot_policy*. This suggests stronger general social reasoning and better use of explicit cultural feedback. However, both models improve under policy prompting, indicating that the method is useful across different model types.

Context-dependent reasoning remains open. The hardest cases are not clearly acceptable or clearly unacceptable behaviors. Instead, they are cases where the answer depends on details that may not be fully specified. A culturally aware model should not only know social norms, but also know when a norm cannot be applied decisively without additional context.

6 Qualitative Error Analysis

Over-acceptance. Some errors occur when models predict *acceptable* for examples that are actually *depends* or *unacceptable*. This may happen when a model focuses on general politeness or positive intent while ignoring culturally specific expectations, such as hierarchy, ritual timing, or respect.

Over-binarization. The *depends* class is often misclassified as either acceptable or unacceptable. This suggests that models may treat SAC as a binary task even when given three labels. This behavior is most visible for Llama-3.1-8B-

Instruct, which makes zero *depends* predictions under `culture_only`.

Incomplete use of cultural policy. Even when a policy is provided, a model may not fully apply it to the specific scenario. A policy describes the relevant norm, but the model must still connect that norm to the details of the interaction. Errors can occur when this reasoning step is weak or when pretrained assumptions override the provided policy.

Ambiguous ground truth. Some *depends* examples are inherently difficult because different annotators may reasonably disagree. This suggests that future versions of the dataset should include multiple annotators and inter-annotator agreement analysis.

7 Limitations and Future Work

Limitations. Our benchmark contains 90 examples, which is useful for controlled analysis but too small for strong per-culture conclusions. The cultural policies are written by a small annotator pool and may reflect English-centric framing. The dataset is also written in English, which may fail to capture language-specific cultural cues. Finally, we evaluate only two models, so broader claims require testing more model families and larger open-weight variants.

Future work. Future work should scale the benchmark to include more cultures, more scenario types, and more native annotators. The *depends* label should also be studied more carefully. One possible direction is to ask models to identify what additional information is needed before making a final judgment. Another direction is to compare human-written policies with automatically retrieved cultural rules, which may make cultural feedback more scalable. Future experiments should also study how many few-shot examples are needed and how to select them effectively.

8 Conclusion

We study cross-cultural social acceptability classification and ask whether human-defined cultural feedback can improve model performance. Using a balanced 90-example benchmark, we evaluate GPT-4.1 mini and Llama-3.1-8B-Instruct under culture-only, policy, and few-shot policy prompting conditions. Our results show that explicit cultural policies improve both models, and few-shot

prompting provides further gains. GPT-4.1 mini achieves the best performance, reaching 0.822 accuracy and 0.814 macro-F1.

At the same time, the *depends* class remains difficult for both models. This shows that culturally aware SAC requires more than applying fixed norms: models must also reason about ambiguity, missing context, and social uncertainty. Overall, our findings suggest that lightweight, human-written cultural feedback is a practical method for improving culturally adaptive language model behavior at inference time.

References

- Maxwell Forbes, Jena D. Hwang, Vered Shwartz, Maarten Sap, and Yejin Choi. 2020. [Social chemistry 101: Learning to reason about social and moral norms](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 653–670, Online. Association for Computational Linguistics.
- Abhinav Rao, Akhila Yerukola, Vishwa Shah, Katharina Reinecke, and Maarten Sap. 2025. [NormAd: A framework for measuring the cultural adaptability of large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2373–2403, Albuquerque, New Mexico. Association for Computational Linguistics.
- Yan Tao, Olga Viberg, Ryan S. Baker, and Rene F. Kizilcec. 2024. [Cultural bias and cultural alignment of large language models](#). *PNAS Nexus*, 3(9):pgae346.